Does Directors' & Officers' Liability Insurance Impact on Internal Wage Disparity?

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ABSTRACT

Providing decent jobs, eradicating poverty and reducing inequality fall under the 17 Sustainable Development Goals of the United Nations. We investigate the impact of Directors' and Officers' liability insurance (D&O liability insurance) on internal wage disparity among Chinese listed firms from 2007 to 2022. Our findings reveal that D&O liability insurance is positively correlated with the wage disparity between executives and ordinary employees. This association remains significant after the robustness and endogeneity tests. Furthermore, executive quality, agency issues, subsidy-bolstered performance, reduced scrutiny, and constraint and compensation committees appear to be plausible underlying mechanisms. Finally, our results indicate that the positive association between D&O liability insurance and internal wage disparity is more pronounced in non-state-owned enterprises, firms facing higher market competition, and labor-intensive enterprises. This study sheds new light on the real effects of D&O liability insurance.

KEYWORDS
Directors' & officers' liability insurance; Wage disparity; Managerial opportunism; Corporate governance

JEL CLASSIFICATION
E24; J31; G22; G34

1. Introduction

Directors' and Officers' (D&O) liability insurance is coverage that companies acquire to protect their directors and managers from legal liabilities incurred while acting on the company's behalf. The existing literature has found that D&O liability insurance can alleviate agency problems within firms (Core 1997), thereby helping firms acquire more long-term debt (Li, Padmanabhan, and Huang 2024), encouraging managers to make decisions that favor long-term benefits, including promoting corporate social responsibility performance (Meng, Wang, and Zhong 2024; Yang et al. 2024), enhancing innovation capability (Wang et al. 2020), and increasing R&D investment (Shi, Sun, and Lyu 2023). However, D&O liability insurance also increases management's moral hazard and reduces managers' accountability (Baker and Griffith 2010; Li, Yang, and Zhu 2022), which reduces firms' investment efficiency (Li and Liao 2014) and increases firms' cost of capital (Lin et al. 2013; Chen, Li, and Zou 2016). Despite these doc-

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umented effects, there is a limited understanding of whether and how D&O liability insurance purchases create externalities in corporate compensation decisions, thereby affecting internal wage disparity.

Providing decent jobs, eradicating poverty, and reducing inequality fall under the 17 Sustainable Development Goals of the United Nations. For listed firms, this translates to providing fair remuneration. While different positions naturally entail varying income levels and executive compensation typically exceeds that of ordinary employees, an excessively wide income gap may negatively impact a company's operational performance. This disparity is also considered a crucial factor influencing social income fairness (Bao, Cheng, and Smith 2020; D'Mello, Kwon, and Toscano 2024). Reducing internal wage disparity has become a significant issue that needs to be addressed urgently and requires firms to play a key role (Kong, Huang, and Ma 2023).

This study empirically tests the causal effect of D&O liability insurance on firms' internal wage disparity. The exploration of this relationship has important theoretical and practical implications. First, although existing literature has explored D&O liability insurance in corporate governance and investment activities (Core 1997; Shi, Sun, and Lyu 2023; Chen, Li, and Zou 2016), its influence mechanism on compensation systems remains underexplored. This study constructs a path analysis framework to examine how D&O liability insurance influences internal wage disparity, filling this gap in the understanding D&O liability insurance mechanisms and compensation incentives (Chen, Ma, and Bu 2014; Kong, Huang, and Ma 2023). Second, China's D&O liability insurance market remains nascent with its regulatory framework still under development (Li, Padmanabhan, and Huang 2024), which differs from mature markets (Yuan, Sun, and Cao 2016). Consequently, the combination of high-coverage insurance and significant pay gaps may generate agency problems and risk accumulation shaped by local institutional characteristics. Exploring the relationship between D&O liability insurance and compensation structures can enhance the understanding of how corporate governance tools function within an evolving institutional framework. Third, clarifying this relationship offers important insights into mitigating managerial moral hazard while simultaneously promoting distributive fairness and enhancing risk management.

The Chinese market offers several compelling research perspectives. Compared with developed capital markets, China's D&O insurance market remains in its early stages, with coverage rates substantially lower than those observed in developed markets (Li, Padmanabhan, and Huang 2024). Additionally, China has experienced increasing income inequality in recent years, with disparities exceeding those in developed economies, potentially impeding economic growth (Kong, Huang, and Ma 2023). These characteristics make Chinese firms suitable for this study.

We posit that D&O liability insurance can affect corporate internal wage disparity in two opposite ways. Additionally, companies purchasing D&O liability insurance may reduce their internal pay disparity. First, by transferring litigation risks to insurers (Wang et al. 2020; Li and Liao 2014), D&O liability insurance motivates them to participate actively in corporate internal compensation management decisions as engaged monitors (Core 2000; Park 2017). Because excessive internal wage disparity represents a source of conflict and litigation risk (Adams 1965; Rouen 2020), curbing such wage disparity is a direct measure for insurers to mitigate risks. Furthermore, the purchase of D&O liability insurance may alleviate internal conflicts and reduce managerial anxiety, enabling a greater focus on corporate development and thereby elevating governance ability (Shi, Sun, and Lyu 2023). Firms with higher governance abilities normally provide stronger incentives for ordinary employees (Rajan, Servaes,

and Zingales 2000), which in turn enhances wage fairness.

By contrast, the theory explains how D&O liability insurance can increase internal wage disparity. D&O liability insurance effectively weakens the legal deterrent effect of litigation against managerial opportunism by transferring risks (Li, Yang, and Zhu 2022). This risk insulation triggers two critical dynamics: moral hazard and empowerment. On the one hand, shielded executives may pursue personal gains (Yuan, Sun, and Cao 2016) by influencing compensation-setting to secure excessive rewards (Bebchuk, Fried, and Walker 2002), ultimately compromising the interests of employees. On the other hand, this risk mitigation alleviates executives' concerns about decision-making consequences, making governance roles more attractive to highly qualified individuals and teams (Core 1997; Li, Yang, and Zhu 2022), who subsequently demand higher compensation as an incentive (Ye, Li, and Zeng 2022). These combined mechanisms consequently widen internal wage disparity.

To test the above opposing views on the impact of D&O liability insurance on corporate internal wage disparity, we conduct an empirical analysis using data from Chinese listed firms from 2007 to 2022. [1] Our independent variable is whether a company purchases D&O liability insurance, while the dependent variable is the natural logarithm of the difference between the average salary of executives and that of ordinary employees. We find a positive association between D&O liability insurance and the disparity between the wages of executives and ordinary employees. These results are consistent with the second hypothesis, indicating that companies purchasing directors', supervisors', and senior managers' liability insurance widen internal wage disparity.

This study introduces several robustness tests to ascertain the validity of our main findings and strengthen our statistical inference. These tests include alternative dependent variable and sample period tests, dynamic effect tests, Heckman two-stage models, propensity score matching tests, and instrumental variables two-stage least squares tests. Taken together, these analyses support the robustness of our results and provide additional evidence consistent with a positive effect of D&O liability insurance on internal wage disparity.

We then aim to provide direct evidence of the channel through which D&O liability insurance affects internal wage disparity. Our findings indicate that purchasing D&O liability insurance can attract high-quality executives and widen internal wage disparity. Moreover, firms with higher agency issues, greater government subsidies, and without compensation committees have wider internal wage disparities when purchasing D&O liability insurance. Additionally, without effective external governance, such as audits by Big4 firms, purchasing D&O liability insurance further increases internal wage disparity.

Finally, we conduct cross-sectional analyses to evaluate the heterogeneity of the widening internal wage disparity associated with D&O liability insurance. First, the impact of purchasing D&O liability insurance on internal wage disparity is more pronounced in highly competitive markets. Second, this impact is diminished for state-

[1] Our sample consists of A-share listed firms from 2007 to 2022 for several reasons. First, we start from 2007 because the Chinese government implemented new accounting standards in that year, ensuring data consistency and comparability across our sample period. Second, A-share listed companies represent the most important enterprises across various industries in China and are subject to strict regulatory oversight and mandatory information disclosure requirements, providing high-quality and transparent financial data essential for our empirical analysis. Third, the extended sample period allows us to capture the evolving dynamics of D&O insurance adoption in China's developing market and provides sufficient statistical power for robust empirical testing. Finally, focusing on A-share firms ensures our findings are representative of China's domestic capital market, which is particularly relevant given the unique institutional characteristics of emerging markets.

owned enterprises (SOEs). Moreover, purchasing D&O liability insurance can widen internal wage disparity more strongly in labor-intensive enterprises.

We further analyze whether the impact on internal wage disparity stems from excessively high CEO salaries or disproportionately low employee wages. Our results reveal that D&O liability insurance positively affects both CEO and employee wages. However, the increase in CEO wages is more pronounced, thereby exacerbating internal pay disparities.

This study contributes to several strands of literature. To the best of our knowledge, this study is the first to examine the effect of D&O liability insurance on internal wage disparity. We provide empirical evidence supporting the managerial opportunism hypothesis, aligning with Meng, Wang, and Zhong (2024), who suggest that D&O liability insurance increases the cost of internal corporate governance. Thus, we add to the literature that this is an important channel through which D&O liability insurance further leads to the expansion of internal wage disparity within firms.

Second, we provide direct evidence of the factors influencing internal wage disparity. Kong, Huang, and Ma (2023) argue that digital transformation significantly widens the pay gap, and Guoyiming Zhu and Zuo (2023) find a significant positive correlation between clan culture and pay disparity within family firms. Furthermore, political promotion incentives (Gaowen Kong and Liu 2023; Bebchuk, Cremers, and Peyer 2011), and operating performance (Ho et al. 2024) may widen the internal wage disparity. Our study provides new evidence that D&O liability insurance can widen internal wage disparity. Although our empirical findings align with the managerial opportunism views, they provide valuable insights into internal wage allocation.

Third, our findings provide important policy implications for addressing income inequality in emerging markets, particularly China, which has an underdeveloped corporate governance system and faces serious wage disparity issues (Li, Yang, and Zhu 2022). Our results suggest that policymakers and regulatory bodies in emerging markets should carefully evaluate whether D&O liability insurance truly contributes to effective corporate governance, given its potential to exacerbate internal wage disparity.

The remainder of this paper is organized as follows: Section 2 presents our hypotheses development. Section 3 describes the data sources and defines the main variables of interest. Section 4 reports our empirical results on the relationship between D&O liability insurance and internal wage disparity, including robustness checks and a discussion of the potential underlying channel. Finally, section 5 concludes the paper.

2. Related research and hypotheses development

2.1. *Theoretical background and related research*

The growing significance of D&O liability insurance has led to an expanding body of literature and theoretical frameworks. Proponents of the monitoring effect hypothesis suggest that D&O liability insurance enhances corporate governance through several mechanisms. First, insurers' scrutiny of client firms strengthens external regulations by effectively monitoring corporate managers (Mayers and Smith Jr 1982; O'Sullivan 1997). Second, it incentivizes executives to prioritize stakeholder interests, resulting in lower agency costs, enhanced investor protection, and reduced corporate misconduct (Core 2000; Li, Yang, and Zhu 2022; Wang and Wu 2023). This insurance also reinforces firms' internal control systems (Priest 1986) and enables independent oversight

of management decisions by third-party insurers (Core 2000). Third, D&O liability insurance enhances managers' risk tolerance and promotes a long-term strategic focus (Chen, Chen, and Yang 2017). Research has documented additional corporate benefits, including reduced negative announcement effects, increased SEO announcement returns (Liao, Chuang, and Wang 2022), and lower stock price crash risk (Yuan, Sun, and Cao 2016).

Conversely, opponents support the managerial opportunism hypothesis (moral hazard hypothesis). They argue that D&O liability insurance's protective effect against nuisance suits may lead to unintended moral hazard and lax management (Baker and Griffith 2010), thereby exacerbating agency problems (Li, Yang, and Zhu 2022). This could result in the infringement of shareholders' interests (Lin et al. 2013). Moreover, as coverage becomes more comprehensive, directors and executives become less risk-averse (O'Sullivan 1997; Boyer and Tennyson 2015), potentially pursuing personal gains at stakeholders' expense (Yuan, Sun, and Cao 2016; Li, Padmanabhan, and Huang 2024). This behavior may manifest as performance manipulation through the disguising of results (Bergstresser and Philippon 2006; Gaowen Kong and Liu 2023), and a greater maturity mismatch (Yunxi Tang and Yao 2023). Finally, D&O liability insurance may weaken the board's accountability system, allowing managers to pursue personal interests with fewer constraints (Baker and Griffith 2010).

The managerial power hypothesis argues that executives leverage their influence to set their own compensation with minimal restrictions by manipulating the board's pay-setting process, a privilege unavailable to ordinary employees (Bebchuk, Fried, and Walker 2002; Bebchuk and Fried 2006; Choe, Tian, and Yin 2014). This increased agency cost widens wage disparities within firms (Bebchuk, Cremers, and Peyer 2011). When firm performance improves, executives often receive disproportionately larger salary increases than replaceable ordinary employees, further expanding the internal pay gap (Gaowen Kong and Liu 2023). Chen, Ma, and Bu (2014) contend that insufficient monitoring and opportunistic management behavior can contribute to internal income inequality.

The existing literature introduces various determinants and theories of internal wage disparity. Although these studies do not explicitly address the direct impact of D&O liability insurance, there are several reasons to believe that the unique role of D&O liability insurance in corporate operations may contribute to widening internal wage gaps.

2.2. *Hypothesis development*

One of the key responsibilities of the board of directors is setting executive compensation and having the authority to dismiss negligent executives (Fama and Jensen 1983). According to the managerial opportunism hypothesis, purchasing D&O liability insurance weakens the accountability mechanism for directors (Baker and Griffith 2010), potentially diminishing their sense of responsibility, as they feel protected by insurance and significantly reducing constraints on management.

The managerial power hypothesis (Bebchuk, Fried, and Walker 2002) suggests that without effective regulations and checks and balances from responsible directors, executives may seek excessive compensation based on personal interests by influencing the pay-setting process through their power (Bebchuk and Fried 2006). Ordinary employees lack such power. The more powerful the executives become, the more they can set their own wages with fewer restrictions (Choe, Tian, and Yin 2014). This leads to

an increase in internal wage disparity.

The substantial wage gap between executives and ordinary employees is likely influenced by executives' pursuit of self-interest (Kong, Huang, and Ma 2023). The managerial opportunism hypothesis suggests that purchasing D&O liability insurance may exacerbate opportunistic behavior, potentially emboldening executives to seize personal gains more brazenly and with impunity, even at the expense of other stakeholders (Yuan, Sun, and Cao 2016). This is because D&O liability insurance provides executives with legal and financial protection, significantly reducing their risk of litigation and potential losses arising from irregular behavior in discharging their duties (Lin et al. 2013; Wang and Chen 2016).

Furthermore, D&O liability insurance, which acts as a safeguard, alleviates executives' concerns about management and decision-making errors. This often attracts high-quality individuals and teams to take on corporate governance roles (Core 1997; Li, Yang, and Zhu 2022). These executives may demand higher compensation as an incentive (Ye, Li, and Zeng 2022), potentially widening the internal wage gap. Based on these arguments, we propose the following hypothesis:

H1. *Ceteris paribus, purchasing D&O liability insurance widens the wage disparity between executives and ordinary employees.*

Bebchuk, Cremers, and Peyer (2011) argue that the agency problem arises from the inherent misalignment between shareholders' and management's interests, with internal wage disparity often resulting from executives receiving excessive compensation in pursuit of their own interests. Chen, Ma, and Bu (2014) and Gaowen Kong and Liu (2023) concur that only robust corporate governance can effectively curb opportunistic behaviors, thereby mitigating internal wage disparity.

D&O liability insurance may trigger moral hazard by reducing managers' incentives to serve stakeholders' best interests (Chung and Wynn 2008; Yuan, Sun, and Cao 2016). Wang and Chen (2016) show that purchasing D&O liability insurance reduces the sensitivity of directors' compensation to performance and increases agency costs. Additionally, the establishment of compensation committees can strengthen the correlation between executive compensation and firm performance (Merhebi et al. 2006) and mitigate excessive CEO pay due to agency conflicts (Conyon and Peck 1998). Consequently, firms with severe agency problems or weak internal governance mechanisms tend to exhibit wider wage gaps when purchasing D&O insurance. Based on these arguments, we propose the following hypothesis.

H2. *The positive relationship between D&O liability insurance purchase and internal wage disparity is stronger in firms with more severe agency problems.*

Executive compensation is often tied to organizational performance, with executives typically receiving larger compensation increases for superior company performance than ordinary employees (Gaowen Kong and Liu 2023). This creates incentives for executives to manipulate firm performance to meet appraisal requirements (Jensen 1986). Executive managerial power may allow managers to present performance bolstered by government subsidies more favorably, which can widen internal wage disparity (Bergstresser and Philippon 2006; Gaowen Kong and Liu 2023). When protected by D&O insurance, executives may be more willing to attribute such subsidy-bolstered performance to managerial ability. Based on this argument, we propose the following hypothesis.

H3. *The positive relationship between D&O liability insurance purchase and internal wage disparity is stronger in firms with greater subsidy-bolstered performance.*

High-quality external corporate governance, such as audits by Big 4 firms, tends to result in stricter audit opinions for companies with substantial D&O liability insurance coverage. This leads to more rigorous control over abnormal profit adjustments and financial goal achievements, thereby inhibiting opportunistic behavior (Chi and Weng 2014). As previously noted, internal wage disparity often widens because of management opportunism. Colpan and Yoshikawa (2012) find that effective governance mechanisms control internal wage disparity resulting from executive power abuse, thereby reducing opportunistic behavior by executives in pursuit of excessive pay and ultimately lowering the firm's overall internal wage disparity. Faleye, Hoitash, and Hoitash (2011) show that firms with stronger oversight have lower CEO pay levels and are less likely to engage in earnings management. Chen, Ma, and Bu (2014) also argue that stricter external governance effectively mitigates firms' internal wage disparity. In other words, a weaker external governance is associated with a wider internal wage gap. Thus, we propose the following hypothesis:

H4. *The positive relationship between D&O liability insurance purchase and internal wage disparity is stronger in firms with weaker external corporate governance.*

D&O liability insurance mitigates these concerns by providing executives with operational safeguards and reducing their exposure to personal litigation risks (Li, Padmanabhan, and Huang 2024). Executives tend to prioritize personal interests and risk aversion in their decision-making (Smith and Stulz 1985). Therefore, as a tool for internal governance, this risk-mitigation mechanism attracts higher-quality executives. Meanwhile, high-quality executives typically demand compensation premiums with higher bargaining power, which may ultimately lead to an expansion in the internal wage disparities within firms (Ye, Li, and Zeng 2022). Hence, we propose the following hypothesis:

H5. *Purchasing D&O liability insurance aggravates internal executives-ordinary employees wage disparity by attracting high-quality executives.*

Compensation committees help mitigate excessively high executive compensation resulting from agency conflicts that deviate from shareholder interests by strengthening pay oversight, restricting managerial discretion, and promoting alignment with shareholder interests (Conyon and Peck 1998; Kanapathippillai et al. 2019). In China, where SOEs account for a relatively high proportion, policies such as the wage restriction order are widely implemented (Xie and Li 2025). By strictly enforcing these institutional constraints, compensation committees can more effectively curb the expansion of internal wage disparity triggered by factors such as D&O insurance (Firth, Fung, and Rui 2006). Based on this, we propose the following Hypothesis:

H6. *The positive relationship between D&O liability insurance purchase and internal wage disparity is stronger in firms without compensation committees.*

3. Research design

3.1. *Data sources*

This study examines A-share listed firms in China from 2007 to 2022, using data from the CSMAR and CNRDS databases. We select 2007 as the starting year because this year marks China's convergence of national accounting standards with International Financial Reporting Standards (IFRS) (Zhang, Andrew, and Rudkin 2012). To ensure

data quality, we follow Gaowen Kong and Liu (2023) in excluding financial sector companies and those labeled as ST or *ST. All continuous variables are winsorized at 1% and 99% to mitigate outlier effects. The final sample comprises 38,440 firm-year observations.

3.2. *Measurement of wage disparity*

Internal wage disparity serves as the dependent variable in our baseline regression model. Following Ye, Li, and Zeng (2022) and Wang and Wu (2023), we measure internal wage disparity using the natural logarithm of the difference between the average executive wage and the average ordinary employee wage, as shown in Eq. (1):

$$Wage\,Disparity_{i,t} = Ln(\sqrt{Executive\,wage_{i,t}} - \sqrt{Employee\,wage_{i,t}}) \tag{1}$$

where $Executives\ wage_{i,t}$ represents the wages of firm i's executives in year t, excluding non-compensated individuals. Given the minimal annual allocation of shares or stock options by Chinese listed companies (Xu, Kong, and Kong 2017; Ye, Li, and Zeng 2022), we exclude equity and stock options from executive compensation.

We follow Kong, Huang, and Ma (2023) for robustness tests, and employ three alternative wage disparity measures: $WageAlter1$, $WageAlter2$ and $WageAlter3$. $WageAlter1$ is the ratio of the average executive wage to the average ordinary employee wage. $WageAlter2$ is the natural logarithm of the difference between the average wage of the top three executives and the average ordinary employee wage. $WageAlter3$ is the natural logarithm of the difference between the average executive wage and the bottom decile of employee wages.

3.3. *Measurement of D&O liability insurance*

We measure our core independent variable, D&O liability insurance, using a dummy variable ($D\&O$). This variable equals 1 if a firm purchases D&O liability insurance in a given year, and 0 otherwise (Zou et al. 2008; Li, Yang, and Zhu 2022; Yuan, Sun, and Cao 2016).

3.4. *Measurement of control variables*

The board structure and the authority of directors exert governance influence on compensation design, thereby affecting wage disparity (Chen, Ma, and Bu 2014). Therefore, this study introduces CEO duality ($Dual$), the largest shareholder's ownership ratio ($Top1$), board size ($Board$), and the proportion of independent directors ($Indep$) to control for the impact of board factors on the model (Chen, Ma, and Bu 2014; Ye, Li, and Zeng 2022). Firm size and age represent the stage of corporate development, affecting job mobility (Liu 2024) and consequently influencing compensation levels for ordinary employees. Additionally, return on assets ratio, financial leverage, and cash flow are crucial metrics for evaluating firm performance (Zhang et al. 2025), and operational performance impacts executive compensation, thereby affecting corporate pay gaps (Ho et al. 2024). Therefore, following the methodology of Kong, Huang, and Ma (2023), we introduce firm age (Age), firm size ($Size$), leverage ratio (Lev), return on assets (Roa), and operating cash flow ($Cashflow$) as control variables. Detailed measurements of all main variables are provided in Table A1 of Appendix A.

3.5. *Econometric specifications*

We employ Ordinary Least Squares (OLS) regression to examine the relationship between D&O liability insurance and internal wage disparity. The choice of OLS is justified by several methodological considerations. First, our theoretical framework posits a linear relationship between D&O insurance purchases and wage disparity, making OLS the most appropriate estimation technique for testing this hypothesis. Second, our dependent variable is continuous, and OLS provides unbiased and efficient estimates under standard assumptions (Wooldridge 2016). Third, the large sample size (38,440 firm-year observations) ensures the asymptotic properties of the OLS estimators. This approach is consistent with the prior literature examining similar relationships in corporate governance research (Wang and Chen 2016; Yuan, Sun, and Cao 2016; Liao, Chuang, and Wang 2022).

The baseline regression specification is:[2]

$$Wage\,Disparity_{i,t} = \beta_0 + \beta_1 D\&O_{i,t} + \beta_2^{'} Controls_{i,t} + \gamma_k + \theta_t + \varepsilon_{i,t} \tag{2}$$

where γ_k and θ_t represent industry- and year-fixed effects, respectively. The vector $Controls_{i,t}$ includes all the other control variables, as defined in Section 3.4. Standard errors are clustered at the firm level to account for potential within-firm correlations in the error terms.

4. Empirical results

This section presents a comprehensive empirical analysis of the relationship between D&O liability insurance purchases and the internal wage disparity. Our baseline regression results constitute the core findings of this study, demonstrating that purchasing D&O liability insurance widens internal wage disparity significantly. To ensure the robustness of these results, we conduct extensive robustness checks using alternative variable measurements, sample periods, Driscoll-Kraay and two-way clustered standard errors and firm-fixed effects, while addressing potential endogeneity concerns through instrumental variables, Heckman two-stage models, propensity score matching, and dynamic effect analyses. These tests consistently support our main findings.

We further explore the underlying mechanisms through economic channel tests to explain how D&O insurance influences wage structures. The cross-sectional heterogeneity tests examine how this relationship varies according to market competition, ownership type, and industry specificity.

[2]Our model incorporates industry and year-fixed effects to control for unobserved heterogeneity that may affect both D&O insurance adoption and wage disparity. Industry fixed effects capture time-invariant industry characteristics that influence corporate governance practices and compensation structures, such as different litigation risks, regulatory requirements, and peer effects within industries. Year-fixed effects control for macroeconomic conditions and regulatory changes that affect all firms simultaneously. This specification helps mitigate omitted-variable bias arising from unobserved factors correlated with the treatment and outcome variables.

We do not include firm fixed effects in our baseline specifications for several reasons. First, our set of time-varying firm-level control variables effectively captures the individual firm heterogeneity that changes over time. Second, the relatively low within-firm variation in D&O insurance purchases limits the identification power of firm fixed effects, potentially leading to imprecise estimates. Third, this approach is consistent with established practice in the corporate governance literature when examining similar research questions (Yuan, Sun, and Cao 2016; Chen, Li, and Zou 2016). Nonetheless, we acknowledge this limitation and address potential endogeneity concerns through various robustness tests, including instrumental variables and propensity score matching approaches, as detailed in subsequent sections.

Finally, the component analysis decomposes wage disparity to determine whether it stems from excessive executive compensation or suppressed employee wages, thus offering deeper insights into the drivers of the observed effect.

4.1. *Descriptive statistics and correlation analysis*

Table 1 presents the descriptive statistics of the main variables. The wage disparity values range from 5.176 to 14.847, with a mean of 12.605 and a standard deviation of 0.902. These statistics align closely with the results reported by Kong, Huang, and Ma (2023). The substantial standard deviation indicates considerable variation in wage disparity across the observations.

The key independent variable, D&O liability insurance, has a mean value of 0.096, indicating that approximately 9% of our firm-year observations indicate D&O liability insurance coverage. This adoption rate aligns with the relatively early stage of development of the D&O liability insurance market in China during the sample period. Regarding the control variables, Age, $Size$, Lev, Roa, $Cashflow$, $Dual$, $Top1$, $Board$, and $Indep$ have mean values of 2.127, 22.180, 0.430, 0.041, 0.048, 0.270, 0.343, 2.131, and 0.375 respectively. The control variables exhibit reasonable distributions, consistent with the findings of prior studies on Chinese listed companies (Ye, Li, and Zeng 2022; Kong, Huang, and Ma 2023).

Table 1. Descriptive statistics.

Variable	N	SD	Mean	Min	Median	Max
Wag disparity	38,440	0.902	12.605	5.176	12.652	14.847
D&O	38,440	0.294	0.096	0.000	0.000	1.000
Age	38,440	0.818	2.127	0.000	2.303	3.497
Size	38,440	1.330	22.180	14.942	21.989	28.636
Lev	38,440	0.207	0.430	0.008	0.424	2.529
Roa	38,440	0.075	0.041	-1.146	0.039	1.285
Cashflow	38,440	0.076	0.048	-0.888	0.047	0.839
Dual	38,440	0.444	0.270	0.000	0.000	1.000
Top1	38,440	0.150	0.343	0.003	0.321	0.900
Board	38,440	0.201	2.131	0.693	2.197	2.890
Indep	38,440	0.056	0.375	0.000	0.357	1.000
Iv_D&O	38,440	0.043	0.096	0.006	0.088	1.000

Notes: This table presents descriptive statistics for firm–year observations of our final sample between 2007 and 2022. D&O liability insurance is an indicator variable that equals 1 if firm purchases D&O liability insurance in a given year and 0 otherwise. The details of the construction of all other variables are described in Table A1.

Table 2 presents Pearson correlation coefficients between variables. The correlation coefficients are relatively small, and D&O is positively correlated with wage disparity, providing initial support for hypothesis H1. The last column shows that the mean Variance Inflation Factor (VIF) is 1.33, which is below the critical value of 10, indicating that multicollinearity is not a concern in our regression analysis.

4.2. *The effect of D&O liability insurance on wage disparity*

We conduct an OLS regression using Eq. (2) to explore the impact of purchasing D&O liability insurance on internal wage disparity. Table 3 presents the regression results under various conditions. All regression coefficients of D&O are positive and significant at the 1% level, thus supporting hypothesis H1. This finding indicates that

Table 2. Pearson correlation analysis.

	Wage Disparity	D&O	Age	Size	Lev	Roa	Cashflow	Dual	Top1	Board	Indep	VIF
Wage Disparity	1.000											
D&O	0.190***	1.000										1.08
Age	0.072***	0.148***	1.000									1.34
Size	0.419***	0.261***	0.383***	1.000								1.66
Lev	0.039***	0.114***	0.349***	0.446***	1.000							1.54
Roa	0.181***	-0.033***	-0.188***	0.030***	-0.347***	1.000						1.40
Cashflow	0.140***	0.021***	-0.008	0.072***	-0.150***	0.392***	1.000					1.20
Dual	0.008	-0.071***	-0.239***	-0.166***	-0.144***	0.032***	-0.011**	1.000				1.10
Top1	-0.012**	0.016***	-0.074***	0.201***	0.048***	0.130***	0.092***	-0.069***	1.000			1.09
Board	0.038***	0.071***	0.133***	0.243***	0.150***	0.021***	0.049***	-0.189***	0.036***	1		1.51
Indep	0.019***	0.018***	-0.010*	0.032***	-0.005	-0.028***	-0.018***	0.107***	0.034***	-0.509***	1.000	1.40

Notes: This table presents the Pearson correlation coefficients between D&O liability insurance, wage disparity, and control variables. *,**, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

purchasing D&O liability insurance widens firms' internal wage disparity, consistent with the managerial opportunism view.

We also conducted a series of additional analyses, including robustness checks, endogeneity tests, economic channel analyses, cross-sectional heterogeneity tests, and component analyses. Owing to space constraints, detailed results are provided in Appendix B.

Our findings align with the concerns raised by Baker and Griffith (2010) and Li, Yang, and Zhu (2022) regarding the moral hazard risks associated with D&O liability insurance. Specifically, our evidence suggests that in the Chinese context, the protective effect of D&O liability insurance appears to outweigh any oversight benefits, leading to increased managerial opportunistic behavior manifested as widening compensation gaps. The low coverage rate of directors' and officers' liability insurance in China (Li, Padmanabhan, and Huang 2024), coupled with high income inequality (Kong, Huang, and Ma 2023), suggests that D&O liability insurance may exacerbate existing governance deficiencies.

5. Conclusions

Despite the increasing popularity of D&O liability insurance, evidence regarding its role in corporate governance remains divergent, primarily focusing on the debate between the managerial opportunism and the monitoring effects hypotheses. Using a sample of 38,440 annual firm observations of Chinese listed companies from 2007 to 2022, our model documents a positive association between purchasing D&O liability insurance and internal wage disparity, while controlling for other variables. Furthermore, our findings suggest that executive quality, agency problems, compensation committees, subsidy-bolstered performance, and lack of scrutiny and constraints may serve as the economic channel mechanisms for this relationship.

The positive relationship between purchasing D&O liability insurance and the internal wage disparity revealed in this study is consistent with the institutional and cultural characteristics of China's transition economy. First, Chinese firms that are facing economic restructuring can use D&O liability insurance as a governance risk buffer while simultaneously widening internal wage disparity to strengthen human capital incentives, reflecting the rebalancing of efficiency and risk in the restructuring economy. Second, non-SOEs are not constrained by "salary cap" regulations and are free to determine their compensation structures. They circumvent the equality constraints by covering litigation risks through insurance, reflecting differentiated em-

Table 3. Regression results of D&O liability insurance and wage disparity.

Variables	(1)	(2)	(3)
	Wage Disparity	Wage Disparity	Wage Disparity
D&O	0.583***	0.412***	0.179***
	(0.015)	(0.035)	(0.028)
Age			-0.072***
			(0.012)
Size			0.308***
			(0.009)
Lev			-0.141***
			(0.052)
ROA			1.588***
			(0.106)
Cashflow			0.719***
			(0.086)
Dual			-0.004
			(0.017)
Top1			-0.367***
			(0.069)
Board			0.073
			(0.052)
Indep			-0.279*
			(0.152)
Constant	12.549***	12.565***	5.947***
	(0.005)	(0.011)	(0.210)
Year FE	No	Yes	Yes
Industry FE	No	Yes	Yes
N	38,440	38,440	38,440
adj. R^2	0.036	0.201	0.373

Notes: This table presents the regression results for the impact of purchasing D&O liability insurance on internal wage disparity. Year FE represents time-fixed effects, and Industry FE represents industry-fixed effects. Standard errors in parentheses are clustered by firms. All variables are defined in Table A1. *,**, and *** denote significance at 10%, 5%, and 1% significance levels, respectively.

powerment under ownership separation in China's market governance.

Additional tests strengthen our baseline regression results and clarify the association between purchasing D&O liability insurance and the internal wage disparity. Our heterogeneity analysis reveals that the positive association between D&O liability insurance and wage disparity is more pronounced in firms operating in more competitive markets, non-SOEs enterprises, and labor-intensive industries. The component analysis demonstrates that the widening wage gap primarily results from increased managerial compensation following the adoption of D&O liability insurance, whereas ordinary employee wages remain relatively stable. Overall, these findings support the managerial opportunism hypothesis documented in prior literature.

Given that the widening effect of purchasing D&O liability insurance on internal wage disparity is stronger in weak regulatory environments, regulators should prioritize strengthening oversight of non-SOEs and firms with weak governance foundations. Such firms could be required to provide mandatory and detailed disclosure of D&O liability insurance terms and executive and employee compensation structures. This approach significantly enhances transparency, enables more precise targeting of regulatory resources to areas of risk concentration, and provides the market with critical data to assess potential agency cost increases.

Boards of directors, particularly those of non-state-owned companies and companies without established compensation committees, must subject D&O liability insurance purchase decisions to mandatory joint considerations with executive compensation

program design. Compensation committees should explicitly assess the moral hazard that existing or proposed D&O liability insurance may pose and its potential contribution to internal wage disparity when evaluating the reasonableness of executive compensation. For companies with more pronounced agency issues, the introduction of independent consultants should be considered when conducting specialized assessments of this relationship.

Particularly, investors concerned with governance risk should pay special attention to the coexistence of D&O liability insurance purchases with significant internal wage disparity when evaluating companies, especially if the company lacks effective oversight mechanisms, such as having no compensation committee or operating in a weak regulatory environment. This pattern should be viewed as a potential warning signal of heightened potential agency conflicts and increased management defenses and should be incorporated into risk assessment models and due diligence priorities.

We emphasize that the mechanisms revealed in this study and their policy implications are context dependent. The amplifying effect of D&O liability insurance on internal wage disparity operates primarily and is more pronounced in environments with clear deficiencies in governance mechanisms, where internal checks and balances are weak (severe agency problems, a lack of compensation committees), external constraints are loose, and firms have specific ownership characteristics (non-SOEs). This effect is less evident in firms with mature governance and sound oversight mechanisms or in strictly regulated SOEs. Therefore, policymakers, corporate practitioners, and investors must carefully consider the specific governance structure of the target firm, the nature of its ownership, and the strength of the regulatory environment in which it operates when applying these recommendations. This contextualized and differentiated application approach will greatly enhance the practical guidance value of the research findings and the relevance of policy recommendations.

This study offers several opportunities for future research. First, we used a sample of companies listed in China, the world's largest emerging market. It remains to be determined whether firms from mature or emerging markets would yield similar findings. Second, our study focused on data from listed companies; extending the research to unlisted firms, especially startups offering shares and options to managers, may yield different insights. Finally, there may be undiscovered potential channels linking D&O liability insurance and internal wage disparity, thus providing avenues for future research.

Disclosure statement

No potential conflict of interest was reported by the authors.

Data statement

The data used in this study are clearly identified within the paper and will be made available upon reasonable request.

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Appendix A. Variable Definitions

Table A1. Variable definitions.

Symbol	Measurement
Wage Disparity	Natural logarithm of the difference between average executive compensation and average employee compensation.
WageAlter1	Ratio of average executive wages to average ordinary employee wages.
WageAlter2	Natural logarithm of the difference between average compensation of the top three executives and average ordinary employee compensation.
WageAlter3	Natural logarithm of the difference between average executive wages and bottom decile employee wages.
D&O	An indicator variable equal to one if firm i purchase D&O liability insurance in year t.
Age	Natural logarithm of listed year.
Size	Natural logarithm of total assets.
Lev	Ratio of total liability to total assets.
ROA	Ratio of net profit to total assets.
Cash	Ratio of net cash flow from operations to total assets.
Dual	An indicator variable equal to one if the chairman of the board and CEO of firm i in year t are the same individual, and zero otherwise.
Top1	The proportion of the largest shareholder.
Board	Natural logarithm of the number directors.
Indep	The proportion of the independent directors.
Mfee	Ratio of management expenses to main business income.
Subsidy	Ratio of government subsidies to total revenue.
Big4	An indicator variable equal to one if firm i in year t employs a Big 4 auditor, and zero otherwise.
HHI	The sum of squared market shares of all firms in an industry. Higher values indicate lower market competition.
SOE	An indicator variable equal to one if firm i is a state-owned enterprise, and zero otherwise.
Hiexecutive	An indicator variable equal to 1 if a firm's proportion of executives with overseas education exceeds the annual industry median, 0 otherwise.
Tenure	Natural logarithm of average executive tenure plus 1 of a given firm.
GDP	Natural logarithm of annual gross domestic product of a province.
Regulatory	Natural logarithm of the number of regulatory inquiry letters received by a firm plus 1.
Culture	Natural logarithm of 1 plus the number of Jinshi in each province.
Committee	An indicator variable equal to 1 if a firm has a compensation committee, 0 otherwise.
Laborint	An indicator variable equal to 1 if a firm is labor-intensive, 0 otherwise.
Exewage	Natural logarithm of the average compensation of executives.
Empwage	Natural logarithm of the average ordinary employee wage.
Iv_D&O	Ratio of firms with D&O liability insurance coverage in the province where the firm is located.

Appendix B. Additional Analysis

B.1. *Robust checks*

Our baseline regression suggests the positive association between D&O liability insurance purchases and internal wage disparity. Although these initial findings support the managerial opportunism hypothesis over the monitoring effect hypothesis, further tests are necessary to establish the robustness of these results. Consequently, we conduct several additional analyses in the following sections, including alternative variable tests, exclusion of global-pandemic disruptions, the Heckman two-stage model, dynamic effect tests, propensity score matching, and the instrumental variable approach.

B.1.1. The alternate measures of Wage Disparity

In Eq. (2), we use $Wage\,Disparity$ as the dependent variable, supporting our baseline regression. However, prior studies have employed several alternative measurements of wage disparity. Following Ye, Li, and Zeng (2022), we use $WageAlter1$ and $WageAlter2$ to replace wage disparity in Eq. (2), while maintaining the same control conditions as in the baseline regression.

Moreover, we use the difference between the average executive compensation and the bottom decile wage of employees ($WageAlter3$) as an alternative proxy for the dependent variable.[3]

Columns (1) - (3) of Table B1 present the regression results using these alternative dependent variables. The coefficients of D&O in all columns remain statistically significant and positive at the 1% level. These findings demonstrate that the impact of D&O liability insurance purchases on internal wage disparity remains robust when using alternative dependent variables. The results continue to support the managerial opportunism hypothesis regarding D&O liability insurance purchases.

B.1.2. Replace the sample time interval

The global pandemic, which began in the early 2020s, prompted many firms to implement salary reductions and layoffs to alleviate economic pressures (Acs and Karpman 2020), potentially affecting internal wage disparity. To mitigate the influence of the pandemic on our baseline regression results, we re-estimate our model using only the observations from 2007 to 2019.

Column (4) of Table B1 presents the regression results excluding observations from 2020 to 2022. The coefficient of D&O remains statistically significant and positive at the 1% level. This suggests that our baseline results are robust.

B.1.3. Alternative model specification with firm-fixed effects

Although our baseline model excludes firm fixed effects for methodological reasons discussed in Section 3.5, we recognize that this choice may raise concerns about omitted variable bias from unobserved firm-specific characteristics. To address this, we re-estimate the baseline model with firm-fixed effects as a robustness check to strengthen the credibility of our results.

Column (5) of Table B1 presents the results of this alternative specification. The coefficient for D&O liability insurance remains positive and statistically significant, suggesting that our main finding is robust to the inclusion of firm-fixed effects.

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[3] As China's A-share listed companies do not disclose specific employee wage details, we cannot obtain the bottom decile employee wage directly from the annual reports and databases. Therefore, following the literature, we draw on Song et al. (2018) and Mueller, Ouimet, and Simintzi (2017) to estimate the bottom decile employee wages. Specifically, we use the within-firm lognormal distribution of wages (Song et al. 2018) to estimate the 10th percentile wage as follows:

$$Wage_{10th} = \exp\left(\ln(\text{MeanWage}) - \frac{\sigma^2}{2} + \sigma \times \Phi^{-1}(0.1)\right) \tag{B1}$$

where $MeanWage$ is the observed average employee wage, Φ is the standard deviation of the log wage estimated from industry wage distribution studies, and $\Phi^{-1}(0.1)$ is the 10th percentile of the standard normal distribution. Following Mueller, Ouimet, and Simintzi (2017), we set $\sigma = 0.5$ in the baseline specification and conduct sensitivity analyses using alternative values from 0.3 to 0.7. The indicator $WageAlter3$ is the natural logarithm of the difference between the average executive wages and the bottom decile employee wages. This measure better captures the gap between executive compensation and the bottom decile employee wages, thus providing a more comprehensive measure of internal wage disparity.

Table B1. The results of robustness checks.

	Alternate measures of Wage Disparity			Alternate sample time interval	Firm fixed effect	Driscoll-Kraay estimator	Alternative standard error clustering
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables	WageAlter1	WageAlter2	WageAlter3	Wage Disparity	Wage Disparity	Wage Disparity	Wage Disparity
D&O	0.459***	0.188***	0.184***	0.192***	0.078***	0.078***	0.181***
	(0.175)	(0.028)	(0.023)	(0.043)	(0.026)	(0.021)	(0.028)
Age	-0.304***	-0.080***	-0.048***	-0.081***	-0.081***	-0.081***	-0.066***
	(0.061)	(0.011)	(0.010)	(0.018)	(0.018)	(0.020)	(0.014)
Size	1.089***	0.298***	0.291***	0.309***	0.267***	0.267***	0.310***
	(0.065)	(0.010)	(0.008)	(0.011)	(0.014)	(0.010)	(0.009)
Lev	0.323	-0.119**	-0.104**	-0.100*	-0.090*	-0.090**	-0.108*
	(0.220)	(0.049)	(0.043)	(0.059)	(0.050)	(0.036)	(0.053)
ROA	7.013***	1.958***	2.333***	2.489***	1.373***	1.373***	2.135***
	(0.519)	(0.106)	(0.107)	(0.136)	(0.084)	(0.159)	(0.211)
Cashflow	3.245***	0.773***	0.677***	0.772***	0.201***	0.201***	0.754***
	(0.445)	(0.082)	(0.077)	(0.101)	(0.057)	(0.052)	(0.108)
Dual	0.079	0.008	-0.005	-0.002	-0.021	-0.021***	-0.004
	(0.078)	(0.016)	(0.014)	(0.020)	(0.014)	(0.007)	(0.016)
Top1	-2.311***	-0.547***	-0.295***	-0.380***	-0.298***	-0.298***	-0.374***
	(0.356)	(0.064)	(0.056)	(0.076)	(0.089)	(0.055)	(0.065)
Board	-0.069	0.079	0.051	0.097	0.076	0.076***	0.062
	(0.274)	(0.051)	(0.044)	(0.060)	(0.047)	(0.017)	(0.055)
Indep	-1.625*	-0.349**	-0.185	-0.185	0.189	0.189	-0.251
	(0.847)	(0.156)	(0.147)	(0.185)	(0.132)	(0.113)	(0.173)
Constant	-17.567***	6.437***	6.906***	5.595***	6.692***	6.017***	5.882***
	(1.395)	(0.218)	(0.192)	(0.246)	(0.332)	(0.260)	(0.238)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	No	No	Yes
Firm FE	No	No	No	No	Yes	Yes	No
N	38,440	38,440	38,440	26,823	38,440	38,440	38,440
adj. R^2	0.204	0.428	0.471	0.353	0.761		0.396

Notes: This table presents the results of the robustness tests. Year FE represents time-fixed effects, and Industry FE represents industry-fixed effects. Standard errors in parentheses are clustered by firm in Columns (1)–(5), and clustered by firm and year in Column (7). All variables are defined in Table A1. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

B.1.4. Driscoll-Kraay standard errors

To address the issues of heterogeneity and cross-sectional dependence, following Driscoll and Kraay (1998), we re-estimate our baseline model using Driscoll-Kraay standard errors, which are robust to both heterogeneity and cross-sectional dependence. Moreover, the two-way clustered standard errors at the firm and year levels (Thompson 2011) are robust.

The results in Column (6) of Table B1 are statistically significant and economically meaningful.

B.1.5. Alternative standard error clustering

To ensure the validity and robustness of the baseline regression, we estimate the baseline model using two-way clustered standard errors at the firm and year levels (Thompson 2011). Column (7) of Table B1 shows that the coefficient of $D\&O$ remains statistically significant across all specifications, supporting that the main findings are robust to different assumptions regarding the error correlation structure.

B.2. *Endogeneity*

B.2.1. Instrumental variable approach

To address potential endogeneity concerns in our baseline results, in which firms with larger internal wage disparities might have greater incentives to purchase D&O liability insurance, we employ the instrumental variable approach. Following Shi, Sun, and Lyu (2023), we use the proportion of firms with D&O liability insurance coverage in each province ($Iv_D\&O$) as the instrumental variable and conduct two-stage least squares (2SLS) regression analysis. A higher provincial D&O insurance penetration directly influences corporate purchase decisions through greater accessibility and lower premiums. However, as a macro-level external factor, provincial insurance penetration is unlikely to directly affect internal mechanisms such as compensation design and corporate culture. Moreover, given the low penetration of D&O insurance in China (Li, Padmanabhan, and Huang 2024), provincial coverage rates may be affected by macro-level regulations but cannot reciprocally influence institutional and regulatory structures. To satisfy the exclusion restriction, we control for provincial GDP (GDP), number of regulatory inquiry letters ($Regulatory$) and the regional distribution of Jinshi degrees (the highest degree in ancient China, $Culture$) to account for provincial economic conditions, regulatory enforcement, and corporate culture, and cluster standard errors at both the province and firm levels (Li, Padmanabhan, and Huang 2024).

In Column (1) of Table B2, the coefficient of $Iv_D\&O$ is positive and statistically significant at the 1% level, indicating a strong correlation between the instrumental variable and D&O liability insurance in the first stage. The Kleibergen-Paap F statistic of 86.94 substantially exceeds the critical value of 16.38 for a 10% maximal IV size, indicating that our instrument is sufficiently strong. The under-identification test strongly rejects the null hypothesis that the equation is under-identified.

In Column (2), the coefficient of D&O liability insurance remains statistically significant, consistent with our baseline regression and supporting its robustness. This finding further validates the managerial opportunism hypothesis, suggesting that purchasing D&O liability insurance may weaken corporate governance incentives. The statistically significant results in Columns (3) and (4) provide additional support for

the empirical validity of the instrumental-variable strategy, although the exclusion restriction cannot be tested directly and should be interpreted with caution

However, we acknowledge an important limitation: the exclusion restriction may not hold perfectly if unobserved province-level factors — such as governance quality, labor market conditions, or enforcement intensity — affect both insurance penetration and wage dispersion at the same time. Although we control for provincial GDP, regulatory inquiry letters, and historical cultural proxies, these measures may not fully address the issue. As a result, some residual confounding may remain, and the IV estimates should be interpreted with caution.

The IV estimate (1.863) is much larger than both the baseline OLS estimate (0.179) and the firm-fixed-effects estimate (0.078). This gap may reflect attenuation from measurement error and the LATE nature of the IV estimate, but it may also indicate residual exclusion-restriction concerns or unobserved selection. Accordingly, the IV estimate should be viewed as evidence of a stronger local effect among compliers, rather than as a precise average effect for all firms. Overall, the results suggest a positive effect of D&O liability insurance on internal wage disparity, although the magnitude remains uncertain and plausibly lies between the conservative firm-fixed-effects estimate and the larger IV estimate. Economically, a one-standard-deviation increase in provincial penetration, 0.043, raises the probability of D&O adoption by 0.040, implying a 0.075-log-point increase in the wage-gap measure, or about a 7.7% increase in the underlying transformed wage-gap measure.[4] This magnitude should be interpreted cautiously, but it suggests that D&O liability insurance may have meaningful implications for internal compensation structures.

B.2.2. Heckman two-stage model

The decision to purchase D&O liability insurance may lead to self-selection bias among firms. To address the potential endogeneity problem arising from selection bias, we employ the Heckman two-stage model (Datta, Doan, and Toscano 2021).

Following Li, Padmanabhan, and Huang (2024), we adopt this model, using the percentage of firms within a certain industry with D&O liability insurance coverage (*per_D&O*) as a variable. In the first stage, we estimate the probability of a firm having D&O liability insurance using a conditional logistic model. We then obtain the Inverse Mills Ratio (IMR) and incorporate it into our baseline model for verification analysis.

Columns (1) and (2) of Table B3 present our findings. The results indicate a significant positive association between D&O liability insurance and internal wage disparity, reinforcing our primary results after controlling for self-selection bias.

The Heckman exclusion variable may also raise an exclusion-restriction concern, as it may capture unobserved industry characteristics related to internal wage disparity. Therefore, the Heckman results should be interpreted as supplementary evidence rather than definitive causal identification.

B.2.3. Control firm quality

To address the potential endogeneity between firm quality, D&O insurance adoption, and executive characteristics, we employ two additional variables to control for firm quality for robustness testing: Tobin's Q ratio and firm growth (Lang and Stulz 1994;

[4]The percentage change is calculated as $\exp(0.040 * 1.863) - 1 = 0.0774$, where 0.040 is the first-stage change and 1.863 is the IV estimate.

Table B2. The results of the instrumental variable approach.

	IV First-stage		IV Second-stage	
	(1)	(2)	(3)	(4)
Variables	D&O	D&O	Wage Disparity	Wage Disparity
Iv_D&O	0.926***	0.936***		
	(0.099)	(0.104)		
D&O			1.863***	1.896**
			(0.301)	(0.735)
Age	0.020***	0.019***	-0.102***	-0.092***
	(0.005)	(0.006)	(0.015)	(0.016)
Size	0.045***	0.046***	0.227***	0.224***
	(0.005)	(0.006)	(0.019)	(0.045)
Lev	-0.007	-0.013	-0.113*	-0.059
	(0.021)	(0.020)	(0.061)	(0.067)
ROA	-0.120***	-0.175***	1.823***	2.424***
	(0.033)	(0.041)	(0.127)	(0.190)
Cashflow	0.061**	0.078**	0.614***	0.608***
	(0.029)	(0.033)	(0.100)	(0.090)
Dual	-0.022***	-0.023***	0.032	0.033
	(0.005)	(0.007)	(0.020)	(0.028)
Top1	-0.035	-0.031*	-0.315***	-0.335***
	(0.025)	(0.016)	(0.082)	(0.074)
Board	0.054***	0.059**	-0.015	-0.039
	(0.020)	(0.028)	(0.065)	(0.068)
Indep	0.117*	0.150**	-0.499***	-0.511**
	(0.063)	(0.076)	(0.191)	(0.199)
GDP		0.002		-0.003
		(0.003)		(0.007)
Culture		0.003		0.007
		(0.002)		(0.010)
Regulatory		0.002		-0.045**
		(0.006)		(0.018)
Kleibergen-Paap F	57.32	80.67		
Under-identification test (p-value)	0.000***	0.017**		
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Clustered	Firm	Firm & province	Firm	Firm & province
N	38,440	38,440	38,440	38,440
adj. R^2	0.132	0.133	0.373	0.395

Notes: This table presents the results of the instrumental variable approach. Year FE represents time fixed effects, and Industry FE represents industry fixed effects. All variables are defined in Table A1. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Hitt, Ireland, and Lee 2000).

Column (3) of Table B3 presents the results of these tests. The coefficient for D&O liability insurance remains positive and statistically significant after controlling for firm quality.

Table B3. The results of the Heckman two-stage approach and control firm quality.

	Heckman First-stage	Heckman Second-stage	Control firm quality
	(1)	(2)	(3)
Variables	D&O	Wage Disparity	Wage Disparity
per_D&O	3.868***		
	(0.337)		
D&O		0.181***	0.179***
		(0.028)	(0.028)
Age	0.162***	-0.067***	-0.068***
	(0.036)	(0.013)	(0.011)
Size	0.260***	0.309***	0.314***
	(0.029)	(0.013)	(0.010)
Lev	0.023	-0.103**	-0.108**
	(0.153)	(0.051)	(0.050)
ROA	-1.155***	2.144***	2.110***
	(0.281)	(0.118)	(0.109)
Cashflow	0.518**	0.758***	0.757***
	(0.234)	(0.089)	(0.086)
Dual	-0.173***	-0.004	-0.005
	(0.046)	(0.017)	(0.016)
Top1	-0.175	-0.368***	-0.371***
	(0.176)	(0.066)	(0.066)
Board	0.379***	0.066	0.062
	(0.134)	(0.052)	(0.051)
Indep	0.857**	-0.258	-0.262*
	(0.426)	(0.159)	(0.156)
Imr		-0.001	
		(0.044)	
TobinQ			0.008**
			(0.004)
Growth			-0.000***
			(0.000)
Constant	-8.807***	5.895***	5.781***
	(0.770)	(0.396)	(0.218)
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
N	38,440	38,440	38,440
adj. R^2	0.185	0.394	0.395

Notes: This table presents the results of the Heckman two-stage approach and control firm quality. Year FE represents time-fixed effects, and Industry FE represents industry-fixed effects. Standard errors in parentheses are clustered by firms. All variables are defined in Table A1. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

B.2.4. Propensity score matching method

We address the potential endogeneity concerns between D&O insurance purchase decisions and internal wage disparity by comparing insured firms with matched control firms on their propensity to purchase D&O insurance. We employ propensity score matching (PSM) following a standard two-stage procedure. First, we estimate a logit model to generate propensity scores for each firm's likelihood of purchasing D&O liability insurance. The logit model includes all the control variables from our baseline regression model in Eq. (2). Second, we use 1:1 nearest neighbor matching without

replacement to select the firms with the closest propensity scores as controls, yielding 3,667 unique treatment firms and 3,667 unique control firms.

Panel A of Table B4 reports our PSM results, showing that the coefficient of D&O liability insurance remains positive and statistically significant. These findings support our main results and the managerial opportunism hypothesis.

Panel B of Table B4 reports the results of covariate balance checks on the mean and median differences in covariates between D&O insured firms and matched control firms using propensity score matching. The results show that most covariates achieve good balance between treatment and control groups after matching, indicating valid matching.

B.2.5. Dynamic effect tests

In previous sections, we established that purchasing D&O liability insurance may widen internal wage disparity. To examine the long-term effects of this insurance on firms, we investigate its dynamic impact on internal wage disparity using alternative dependent variables: $Wage\,Disparity$ at times t+1, t+2, and t+3.

Table B5 presents the results of the dynamic effect tests. Columns (1), (2), and (3) present the coefficients of $Wage\,Disparity$ at times t+1, t+2, and t+3, respectively. All coefficients remain statistically significant and positive at the 1% level, which is consistent with our baseline regression model results. These findings indicate that purchasing (or holding) D&O liability insurance widens internal wage disparity in the long term.

B.3. *Economics channel test*

B.3.1. Agency issues channel

In Section 2.2, we developed hypothesis H2, which examines how purchasing D&O liability insurance may increase agency problems, further exacerbating internal wage disparity. Following Danlu Bu and Lin (2016), we use the management expense rate (management expenses divided by operating income, denoted as $Mfee$ to measure agency costs) in a given firm. We conduct an OLS regression with an interactive multiplication term in Eq. (2) to further understand how D&O liability insurance affects internal wage disparity.

Column (1) of Table B6 presents the results. The coefficient of $Mfee * Big4$ on $Wage\,Disparity$ is 0.477, which is significantly positive at the 1% level. This indicates that agency issues amplify the exacerbating effect of holding D&O liability insurance on internal wage disparity. This finding directly supports the managerial opportunism hypothesis and aligns with the research of Bebchuk, Cremers, and Peyer (2011), who noted that agency problems lead to excessive executive compensation. When governance is weak and agency costs are high, directors' and officers' liability insurance further undermines accountability structures (Baker and Griffith 2010). Chinese firms undergoing economic transformation face increased agency problems (Lin et al. 2020), and D&O liability insurance may exacerbate such issues.

B.3.2. Performance bolstered by subsidies channel

Hypothesis H3 tests whether purchasing D&O liability insurance increases internal wage disparity by subsidy-bolstered performance. We re-estimate our regression model

Table B4. The results of the propensity score matching approach and balance tests.

Panel A: The results of the propensity score matching approach

	logit regression	OLS regression
	(1)	(2)
Variables	D&O	Wage Disparity
D&O		0.196***
		(0.030)
Age	0.304***	-0.071***
	(0.072)	(0.021)
Size	0.514***	0.292***
	(0.056)	(0.015)
Lev	-0.041	-0.157*
	(0.302)	(0.086)
Roa	-2.383***	2.253***
	(0.559)	(0.208)
Cashflow	1.114**	0.540***
	(0.480)	(0.176)
Dual	-0.351***	0.023
	(0.089)	(0.032)
Top1	-0.364	-0.461***
	(0.333)	(0.116)
Board	0.675***	0.013
	(0.253)	(0.089)
Indep	1.394*	-0.882***
	(0.791)	(0.270)
Constant	-15.980***	6.775***
	(1.471)	(0.341)
Year FE	Yes	Yes
Industry FE	Yes	Yes
N	38,440	6,052
Pseudo R^2	0.173	
adj. R^2		0.416

Panel B: Balance tests

	Treat (D&O = 1)			Control (D&O = 0)			Differences	
Variable	N	Mean	Median	N	Mean	Median	t value	z value
Age	3,667	2.496	0.872	3,667	2.508	0.853	-1.24	0.215
Size	3,667	23.201	1.045	3,667	23.163	1.051	1.07	0.284
Lev	3,667	0.501	0.551	3,667	0.503	0.534	-0.43	0.664
ROA	3,667	0.033	0.034	3,667	0.032	0.036	0.54	0.587
Cashflow	3,667	0.053	0.262	3,667	0.051	0.264	0.87	0.382
Dual	3,667	0.173	0.378	3,667	0.174	0.379	-0.12	0.902
Top1	3,667	0.350	0.477	3,667	0.349	0.477	0.26	0.794
Board	3,667	2.175	0.184	3,667	2.176	0.184	-0.27	0.789
Indep	3,667	0.378	0.054	3,667	0.377	0.055	0.26	0.796

Notes: This table presents the results of the propensity score matching approach. Panel A shows the results of PSM regressions. Panel B reports the balance test results of matching variables. Year FE represents time-fixed effects, and Industry FE represents industry-fixed effects. Standard errors in parentheses are clustered by firms. All variables are defined in Table A1. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table B5. The results of the dynamic effect tests.

	Dynamic effect		
	(1)	(2)	(3)
Variables	Wage Disparity$_{t+1}$	Wage Disparity$_{t+2}$	Wage Disparity$_{t+3}$
D&O	0.171***	0.151***	0.135***
	(0.032)	(0.037)	(0.042)
Age	-0.046***	-0.029**	-0.014
	(0.012)	(0.013)	(0.014)
Size	0.301***	0.293***	0.279***
	(0.010)	(0.010)	(0.011)
Lev	-0.064	-0.076	-0.048
	(0.054)	(0.058)	(0.061)
ROA	2.328***	1.970***	1.901***
	(0.121)	(0.129)	(0.140)
Cashflow	0.920***	1.006***	0.983***
	(0.091)	(0.097)	(0.102)
Dual	-0.010	0.003	0.005
	(0.018)	(0.019)	(0.021)
Top1	-0.377***	-0.364***	-0.350***
	(0.068)	(0.072)	(0.075)
Board	0.026	0.023	0.020
	(0.054)	(0.057)	(0.059)
Indep	-0.341**	-0.246	-0.272
	(0.166)	(0.175)	(0.187)
Constant	6.147***	6.315***	6.653***
	(0.222)	(0.235)	(0.248)
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
N	32,906	28,797	25,387
adj. R^2	0.373	0.336	0.303

Notes: This table presents the results of the dynamic effect. Year FE represents time-fixed effects, and Industry FE represents industry-fixed effects. Standard errors in parentheses are clustered by firms. All variables are defined in Table A1. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

by incorporating measures of subsidy-bolstered performance within firms. Following Shi, Sun, and Lyu (2023), we use the ratio of government subsidies to total revenue (*Subsidy*) to proxy for the extent to which firm performance is supported by government subsidies. We employ an interactive multiplication term of *Subsidy* and *D&O* to test this hypothesis.

Column (2) of Table B6 presents the results. The estimated coefficient of $D\&O * Subsidy$ on $Wage\,Disparity$ is 0.075 and is significantly positive at the 10% level. This indicates that subsidy-bolstered performance reinforces the exacerbating effect of holding D&O liability insurance on firms' internal wage disparity. This finding reflects how D&O liability insurance provides protection for executives, thereby reinforcing their intrinsic motivation, as cautioned by Jensen (1986), to present subsidy-bolstered performance more favorably (Gaowen Kong and Liu 2023). This enables executives to achieve business performance metrics, ultimately exacerbating internal pay gaps within corporations.

B.3.3. Scrutiny and constraint channel

We then test hypothesis H4, which considers the scrutiny and constraint effect on the association between D&O liability insurance and internal wage disparity. We employ *Big4* as a moderator variable, utilizing the interaction term between *Big4* and *D&O* to examine its impact on Eq. (2). The variable Big4 is a dummy variable, with a value of 1 indicating the use of a Big 4 auditor and 0 otherwise.

Column (3) of Table B6 reports the regression results. The coefficient of $D\&O * Big4$ on $Wage\,Disparity$ is -0.188, which is significantly negative at the 5% level. This indicates that scrutiny and constraint significantly reduce the positive association between holding D&O liability insurance and internal wage disparity. This suggests that high-quality external governance mitigates internal wage disparity by curbing managerial opportunistic behavior through stricter oversight (Chi and Weng 2014; Colpan and Yoshikawa 2012; Chen, Ma, and Bu 2014). In emerging markets, where some oversight mechanisms are developing, external supervision plays a key role in corporate governance.

B.3.4. Executive quality channel

In this section, we explore hypothesis H5, which states that purchasing D&O liability insurance attracts more highly qualified executives and thus increases internal wage disparity. Following Xie and Li (2025), we employ executives with overseas education backgrounds (*Hiexecutive*) to measure high-quality executives. Executive tenure typically signifies stronger trust and managerial capability (Chen and Fan 2025). Therefore, we also employ executive tenure (*Tenure*) as a robust measure of high-quality executives. Specifically, a value of 1 is assigned to the firm if the percentage of executives with overseas education exceeds the annual industry median, and 0 otherwise. The same applies to executive tenure.

Columns (4) and (5) of Table B6 show that the coefficients of$D\&O * Hiexecutive$ and $D\&O * Tenure$ on $Wage\,Disparity$ are both significantly positive at the 5% level. This indicates that purchasing D&O liability insurance may attract high-quality executives,thereby widening internal wage disparity. Core (1997) and Li, Yang, and Zhu (2022) note that D&O liability insurance helps attract talented executives. Our findings extend their research by showing that this attraction amplifies internal inequality. In China's transitioning market, competition for scarce talent is intensifying, and this

dynamic may make society more receptive to such premiums.

B.3.5. Compensation Committee channel

This section tests the mechanism proposed in hypothesis H6, which states that purchasing D&O liability insurance has less impact on internal wage disparity in firms with compensation committees. We introduce the dummy variable *Committee*, where *Committee* equals 1 if the firm has a compensation committee and 0 otherwise.

Column (6) of Table B6 shows that the coefficient of the $D\&O * Committee$ on $Wage\,Disparity$ is -0.079, suggesting that compensation committees mitigate the wage disparity-enhancing effect of D&O insurance. Although statistically marginal, the economic relevance of this moderating role is consistent with evidence from Kanapathippillai et al. (2019), indicating that in transition economies such as China, formal governance mechanisms can serve meaningful economic functions even when statistical evidence is only moderately strong.

B.4. *Cross-sectional heterogeneity*

B.4.1. Market competition

Intense market competition presents executives with greater risks and implicit incentives (Cuñat and Guadalupe 2004). Additionally, this may reduce firms' relative bargaining power, enabling managers to extract excess returns from enterprises (Bebchuk, Fried, and Walker 2002). Therefore, we anticipate that the exacerbating effect of D&O liabilities insurance purchases on internal wage disparities is more pronounced in highly competitive settings. Following Zheng et al. (2021), we employ the Herfindahl-Hirschman Index (HHI, the higher HHI, the lower market competition.) to measure market competition. We create a binary variable equal to 1 for firms with below-average HHI (above-average competition) and 0 otherwise, and test using the interaction term $HHI * D\&O$.

Column (1) of Table B7 shows that the impact of D&O liability insurance purchases on internal wage disparity is more significant in highly competitive markets. Consistent with existing literature, intense market competition provides implicit incentives to executives, weakening firms' relative bargaining power (Cuñat and Guadalupe 2004) and enabling executives to secure compensation premiums (Bebchuk, Fried, and Walker 2002). Ultimately, this amplifies the effect of D&O liability insurance on internal wage disparity. As the world's largest manufacturing nation, competitive pressures stimulate corporate vitality but also readily compound with imperfect internal governance mechanisms, ultimately creating contradictions. Therefore, it is crucial to prevent vicious competition in talent recruitment.

B.4.2. Ownership attribute

China's SOEs have implemented salary caps for executives (Lin et al. 2020), potentially causing a significant decrease in internal wage disparity in SOEs. A study by Kong, Kong, and Lu (2020) is consistent with this view. To investigate whether ownership attributes affect the impact of D&O liability insurance purchases, we introduced a binary indicator variable, SOE. This variable is assigned a value of 1 for SOEs and 0 for non-SOEs. Then, we incorporated an interaction term between the SOE indicator and D&O liability insurance.

Table B6. The results of the economic channel test.

	Agency issues	Subsidy-bolstered performance	Scrutiny and constraint	Executive quality	Executive tenure	Compensation committee
	(1)	(2)	(3)	(4)	(5)	(6)
Variables	Wage Disparity	Wage Disparity	Wage Disparity	Wage Disparity	Wage Disparity	Wage Disparity
D&O	0.190***	0.160***	0.180***	0.159***	0.163***	0.217***
	(0.013)	(0.031)	(0.030)	(0.031)	(0.030)	(0.043)
D&O*Mfee	0.477***					
	(0.223)					
Mfee	0.647***					
	(0.071)					
D&O*Subsidy		0.075*				
		(0.044)				
Subsidy		-0.006				
		(0.015)				
D&O*Big4			-0.188**			
			(0.078)			
Big4			0.246***			
			(0.053)			
D&O* Hiexecutive				0.116**		
				(0.053)		
Hiexecutive				-0.049**		
				(0.020)		
D&O*Tenure					0.084***	
					(0.041)	
Tenure					0.017***	
					(0.005)	
D&O*Committee						-0.079*
						(0.044)
Committee						0.097***
						(0.017)
Age	-0.070***	-0.072***	-0.068***	-0.083***	-0.074***	-0.068***
	(0.005)	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)
Size	0.317***	0.307***	0.295***	0.308***	0.308***	0.306***
	(0.004)	(0.010)	(0.010)	(0.010)	(0.009)	(0.009)
Lev	-0.055**	-0.145***	-0.131**	-0.147***	-0.098*	-0.137***
	(0.025)	(0.053)	(0.053)	(0.052)	(0.050)	(0.051)
ROA	2.328***	1.676***	1.695***	1.601***	2.113***	1.587***
	(0.077)	(0.108)	(0.107)	(0.106)	(0.108)	(0.106)
Cashflow	0.774***	0.707***	0.685***	0.742***	0.717***	0.713***
	(0.061)	(0.086)	(0.085)	(0.086)	(0.085)	(0.085)
Dual	-0.006	-0.007	-0.006	-0.000	-0.007	-0.006
	(0.008)	(0.017)	(0.017)	(0.017)	(0.016)	(0.017)
Top1	-0.351***	-0.395***	-0.409***	-0.391***	-0.364***	-0.364***
	(0.026)	(0.0699)	(0.069)	(0.070)	(0.066)	(0.069)
Board	0.062***	0.078	0.075	0.065	0.061	0.072
	(0.023)	(0.053)	(0.053)	(0.052)	(0.051)	(0.052)
Indep	-0.284***	-0.246	-0.275*	-0.280*	-0.244	-0.305**
	(0.078)	(0.154)	(0.153)	(0.152)	(0.156)	(0.152)
Constant	5.642***	5.959***	6.224***	6.013***	5.873***	5.958***
	(0.536)	(0.319)	(0.194)	(0.212)	(0.209)	(0.210)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
N	38,440	38,440	38,440	38,440	38,440	38,440
adj. R^2	0.398	0.374	0.376	0.373	0.3975	0.375

Notes: This table presents the results of the economic channel test. Year FE represents time-fixed effects, and Industry FE represents industry-fixed effects. Standard errors in parentheses are clustered by firms. All variables are defined in Table A1. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Column (2) of Table B7 indicates that the impact of purchasing D&O liability insurance on internal wage disparity is more pronounced for non-SOEs. This finding aligns with the view of Lin et al. (2020) that wage restriction rules mitigate income inequality. In markets dominated by SOEs, the wage restriction order can correct income disparities. Private enterprises may also adopt similar governance mechanisms, such as coordinating wage caps through industry associations to address governance issues.

B.4.3. Industry specificity

The labor-intensive industries normally employ a large number of low-skilled workers with greater substitutability, low bargaining power, and relatively lower wages that widen the baseline pay gap within the firm (Kong et al. 2024). A high proportion of low-cost labor reduces the company's average ordinary wage, further widening the internal wage disparity. Therefore, we anticipate that the exacerbating effect of D&O liability insurance purchases on internal wage disparities is stronger in labor-intensive industries. We introduce the binary indicator variable *Laborint*. This variable is assigned a value of 1 for labor-intensive firms and 0 for other firms. We then add an interaction term between the labor-intensive firms indicator and D&O liability insurance.

Column (3) of Table B7 indicates that for labor-intensive firms, the effect of purchasing D&O liability insurance on internal wage disparity is more pronounced. This directly reflects low-skilled workers' vulnerability (Kong et al. 2024) and implies that corporate governance mechanisms, such as D&O insurance can have unequal, necessitating context-specific assessments, particularly in industries reliant on low-skilled labor.

B.5. *Component analysis*

To determine whether the widening wage disparity results from excessive executive pay or low general employee wages, we re-examine the impact of D&O liability insurance on executives and ordinary employee wages separately. We use the variables $ExeWage$ and $EmpWage$ to replace the dependent variable in Eq. (2) for the regression. $ExeWage$ and $EmpWage$ are the natural logarithms of the average executive wage and average ordinary employee wage, respectively.

Columns (1) and (2) of Table B8 present regression results showing a significant positive association between purchasing D&O liability insurance and both executive and ordinary employee wages. However, the coefficient for executive wage is stronger than that for ordinary employee wages. This finding suggests that D&O liability insurance promotes the expansion of internal wage disparity primarily by significantly increasing executive income, while its impact on employee wage is less pronounced.

Table B7. The results of cross-sectional heterogeneity.

	Market competition	Ownership attribute	Industry specific
	(1)	(2)	(3)
Variables	Wage Disparity	Wage Disparity	Wage Disparity
D&O	0.261***	0.281***	0.121***
	(0.033)	(0.036)	(0.041)
HHI*D&O	-0.152***		
	(0.040)		
HHI	0.014**		
	(0.014)		
SOE*D&O		-0.164***	
		(0.051)	
SOE		-0.037	
		(0.024)	
Laborint*DO			0.194***
			(0.053)
Laborint			0.113***
			(0.019)
Age	-0.055***	-0.057***	-0.054***
	(0.012)	(0.012)	(0.012)
Size	0.311***	0.312***	0.287***
	(0.009)	(0.009)	(0.010)
Lev	-0.103*	-0.105**	-0.080
	(0.050)	(0.050)	(0.052)
ROA	2.123***	2.126***	2.379***
	(0.108)	(0.108)	(0.116)
Cashflow	0.750***	0.745***	0.603***
	(0.086)	(0.086)	(0.094)
Dual	-0.011	-0.011	0.017
	(0.016)	(0.016)	(0.016)
Top1	-0.340***	-0.338***	-0.491***
	(0.067)	(0.067)	(0.068)
Board	0.081	0.078	-0.038
	(0.052)	(0.052)	(0.054)
Indep	-0.227	-0.228	-0.302*
	(0.156)	(0.156)	(0.165)
Constant	5.787***	5.781***	6.560***
	(0.213)	(0.213)	(0.225)
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	No
N	38,440	38,440	38,440
adj. R^2	0.397	0.397	0.358

Notes: This table presents the results of cross-sectional heterogeneity. Year FE represents time-fixed effects, and Industry FE represents industry-fixed effects. Standard errors in parentheses are clustered by firms. All variables are defined in Table A1. Column (3) presents an industry-specific regression model without industry-fixed effects, because *Laborint* already incorporates industry-disaggregated information. Thus, we capture industry heterogeneity through interaction terms. Using industry-fixed effects results in omitted coefficients because of perfect collinearity. *,**, and *** denote significance at 10%, 5%, and 1% significance levels, respectively.

Table B8. The results of component analysis.

	(1)	(2)
Variables	ExeWage	EmpWage
D&O	0.162***	0.108***
	(0.022)	(0.017)
Age	-0.032***	0.032***
	(0.009)	(0.007)
Size	0.260***	0.096***
	(0.007)	(0.006)
Lev	-0.144***	-0.222***
	(0.038)	(0.032)
ROA	1.583***	0.246***
	(0.082)	(0.069)
Cashflow	0.632***	0.166***
	(0.064)	(0.050)
Dual	-0.006	-0.012
	(0.012)	(0.010)
Top1	-0.233***	0.154***
	(0.050)	(0.041)
Board	0.046	0.048
	(0.038)	(0.032)
Indep	-0.142	0.151
	(0.120)	(0.103)
Constant	7.266***	9.212***
	(0.163)	(0.139)
Year FE	Yes	Yes
Industry FE	Yes	Yes
N	38,440	38,440
adj. R^2	0.502	0.521

Notes: This table presents the results of component analysis. Year FE represents time-fixed effects, and Industry FE represents industry-fixed effects. Standard errors in parentheses are clustered by firms. All variables are defined in Table A1. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

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